# Random Variables and Probability Distribution

A random variable is a variable whose value is determined by chance or randomness. In statistics and probability theory, a random variable is used to describe the outcomes of a random experiment or process. Random variables can be either discrete or continuous.

A **random variable** is just a fancy name for the result of some random process that you can *represent with a number*.

- If you're flipping a coin: Let's say Heads = 1 and Tails = 0, that's a random variable.
- Rolling a die? The random variable can be 1, 2, 3, 4, 5, or 6.
- Measuring temperature randomly in a city? You get numbers like 32.5°C, 33.7°C, etc.

**Sample Space:** Sample space is the set of all possible outcomes of a random experiment, denoted by “S”, used to determine the probability of an event, which is a subset of the sample space.

**Types of Random Variables?**

The two main types of random variables are:

**Discrete random variables:** These variables can only take on a finite or countably infinite number of possible values, such as the outcome of rolling a die or the number of students in a class. Discrete random variables are typically represented by integers or whole numbers, and their probability distribution is a discrete probability mass function.

**Continuous random variables:** These variables can take on any value within a certain range, such as the weight or height of a person. Continuous random variables are typically represented by real numbers, and their probability distribution is a continuous probability density function.

**So now the question is:**

> "How do we talk about *how likely* each number is?"

That’s where the **PMF**, **PDF**, and **CDF** come in.

# 1. PMF – Probability Mass Function

(for Discrete Stuff like dice or coins)

Think of the **PMF** as a table that tells you:

> "What's the probability of getting each possible value?"

**Imagine Rolling a Die.**

The probability table will be

| Value | Probability |
|---|---|
| 1 | 1/6 |
| 2 | 1/6 |
| 3 | 1/6 |
| 4 | 1/6 |
| 5 | 1/6 |
| 6 | 1/6 |

That's a PMF! It gives the mass of probability to each specific point.

**Three Rules of PMF:**

1. All probabilities must be ≥ 0.

2. The sum of all probabilities = 1.

3. It only works for discrete values (countable things).

# 2. PDF – Probability Density Function

(for Continuous Stuff like height, weight, or temperature)

Now suppose we're measuring your **exact height**. Could be 5.7128 feet, 5.7129 feet...
It's impossible to list all possible heights because there are **infinitely many**.

Instead of assigning a probability to each exact value, we say:

> "What's the *probability density* around this area of values?"

The **PDF is a curve**. The **area under the curve between two points** tells you the probability of the value falling in that range.

**Key PDF Insight:**

- The probability at a **single point** is **0**.
- But the **area between two points** (like from 5.7 ft to 5.8 ft) can have a real probability like 0.1 or 0.3.
- The **total area under the curve** = 1 (like 100% probability).

So PDF tells you **where the values are more dense**, like where values are more *likely* to happen.

# 3. CDF – Cumulative Distribution Function

(for both Discrete and Continuous)

Now imagine you want to know:

> "What's the probability that I get a number **less than or equal to** X?"

That's what **CDF** answers.

If you're rolling a die and ask:

> "What's the probability of getting ≤ 3?"

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P(1) + P(2) + P(3) = 1/6 + 1/6 + 1/6 = 0.5
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The CDF at value 3 is 0.5. You just **add up** all the PMF or PDF values **up to that point**.

For continuous values, CDF is just the **area under the PDF curve** from the left up to X.

Think of it like a **progress bar**: As X increases, the CDF shows how much total probability has been "accumulated."

## Probability Distributions

A probability distribution is a list of all of the possible outcomes of a random variable along with their corresponding probability values.

- A probability distribution is a function or a list of all possible outcomes of a random variable and their corresponding probabilities. It describes the likelihood of each event in a random experiment.

- For example, if we toss a coin, the possible outcomes are heads or tails, each with a probability of 1/2. We can represent this as a probability distribution with two possible outcomes and their respective probabilities: {heads: 1/2, tails: 1/2}.
- Similarly, if we roll a die, the possible outcomes are numbers 1 to 6, each with an equal probability of 1/6. We can represent this as a probability distribution with six possible outcomes and their respective probabilities: {1: 1/6, 2: 1/6, 3: 1/6, 4: 1/6, 5: 1/6, 6: 1/6}.
- Another example could be rolling two dice and adding their values to obtain the sum. The possible outcomes range from 2 to 12, and each outcome has a different probability of occurring, which can be represented as a probability distribution with different probabilities for each outcome.

**Types of Probability Distributions**

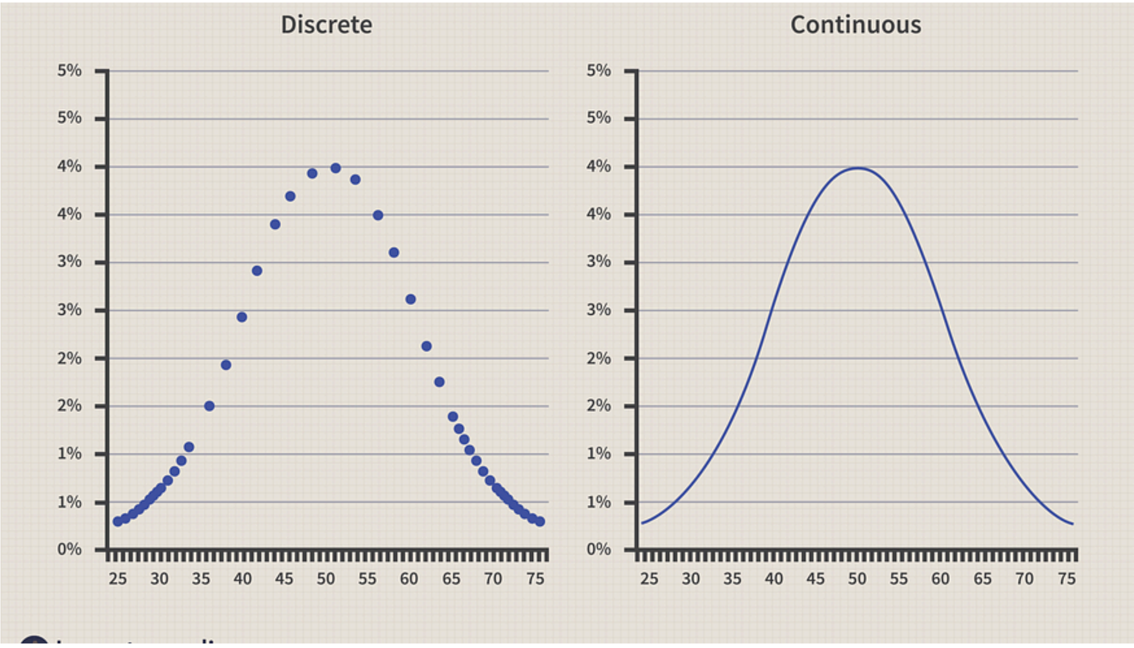
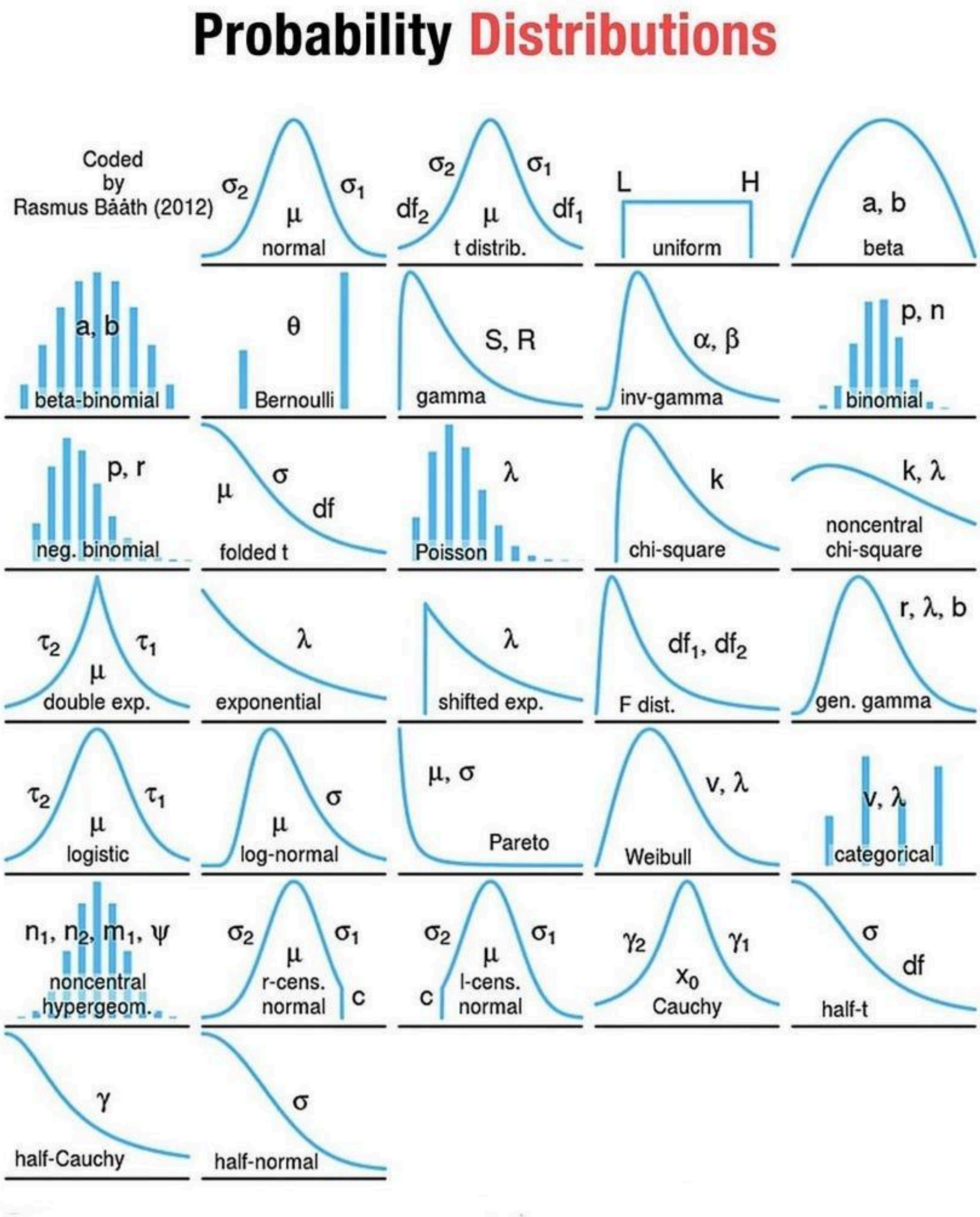



*Fig: Types of Probability Distributions*




**Why are Probability Distributions important?**

- Gives an idea about the shape/distribution of the data.
- And if our data follows a famous distribution then we automatically know a lot about the data.

# Normal Distribution

The normal distribution is a continuous probability distribution that is symmetric about the mean. It forms the famous bell shaped curve.

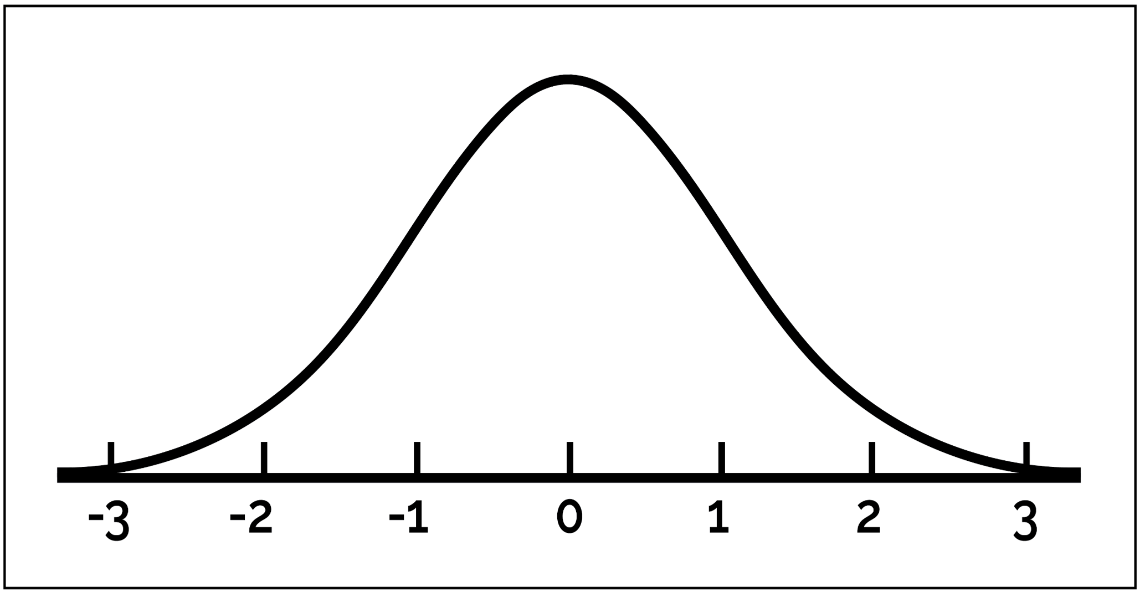



*Fig: Bell Shaped Curve*

Key Features:

1. Symmetric about the mean.
2. Most values are close to the mean.
3. Fewer values as you move away from the mean.
4. Defined entirely by:
    a. Mean
    b. Standard Deviation

## Probability Density Function:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Where:

- x = value for which we want the probability
- μ = mean
- σ = standard deviation
- e = Euler's number (approximately 2.71828)÷
- π = Pi (approximately 3.14159)

## Standard Normal Distribution:

A normal distribution is said to be standard normal distribution if:

1. $\mu = 0$ and,
2. $\sigma = 1$

Then the distribution becomes the **standard normal distribution**, denoted by $Z \sim N(0,1)$. It's PDF is:

f(z) = $\frac{1}{\sqrt{2\pi}} . e^{-z^2/2}$

## Use in Machine Learning

- Naive Bayes: Gaussian NB uses normal distribution for features.
- Anomaly Detection: If a value falls far from the mean (low PDF), it may be an outlier.
- Bayesian Inference: Normal distributions used as priors and posteriors.
- Likelihood estimation: MLE for a normal leads to sample mean and variance.